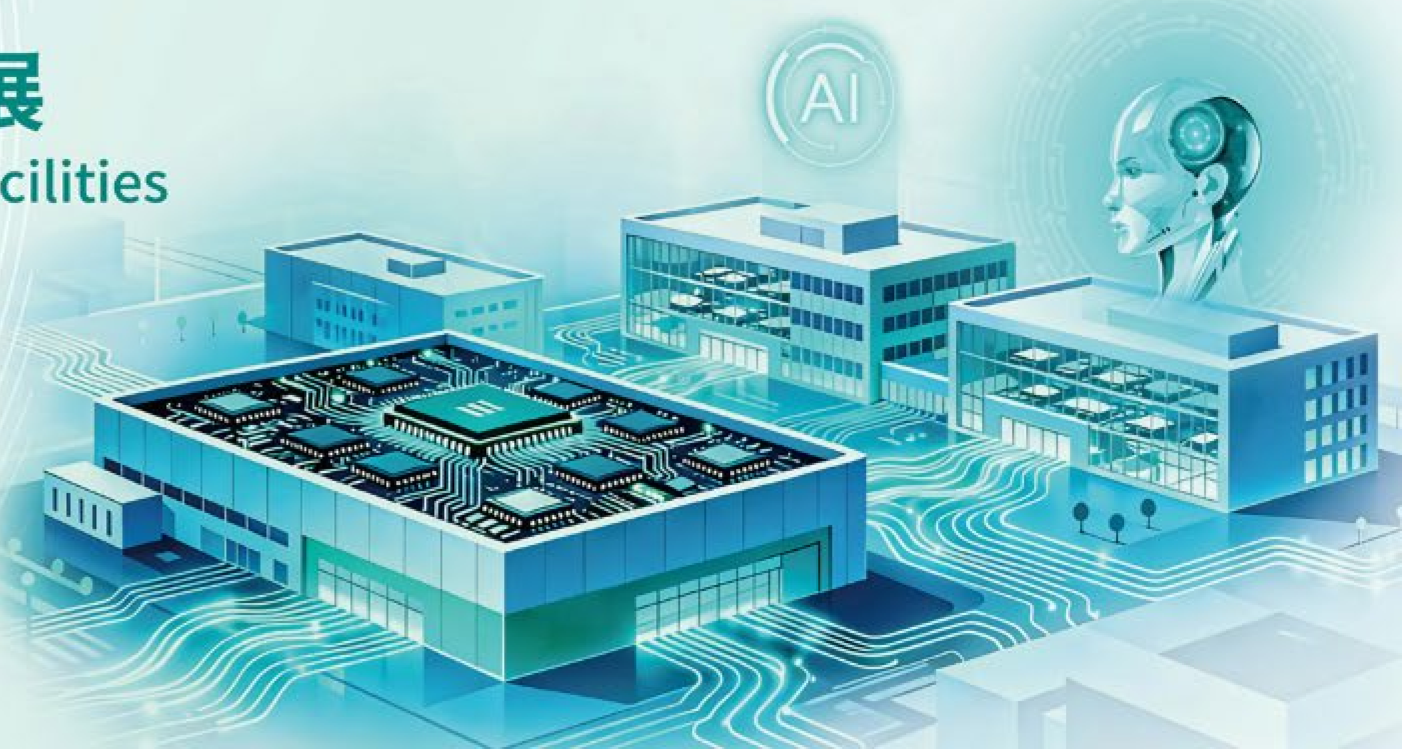
2026高科技廠房設施國際論壇
High-Tech Facility International Forum 2026

以AI驅動創新加速
高科技廠房設施的未來發展

Accelerating the Future of High-Tech Facilities Through AI-Driven Innovation



Thursday, September 3, 2026

創控科技股份有限公司
TRICORNTECH CORPORATION

PHYSICS-CONSTRAINED

AI DIGITAL TWIN

FOR ENVIRONMENTAL RISK PREDICTION IN HIGH-TECH FACILITIES

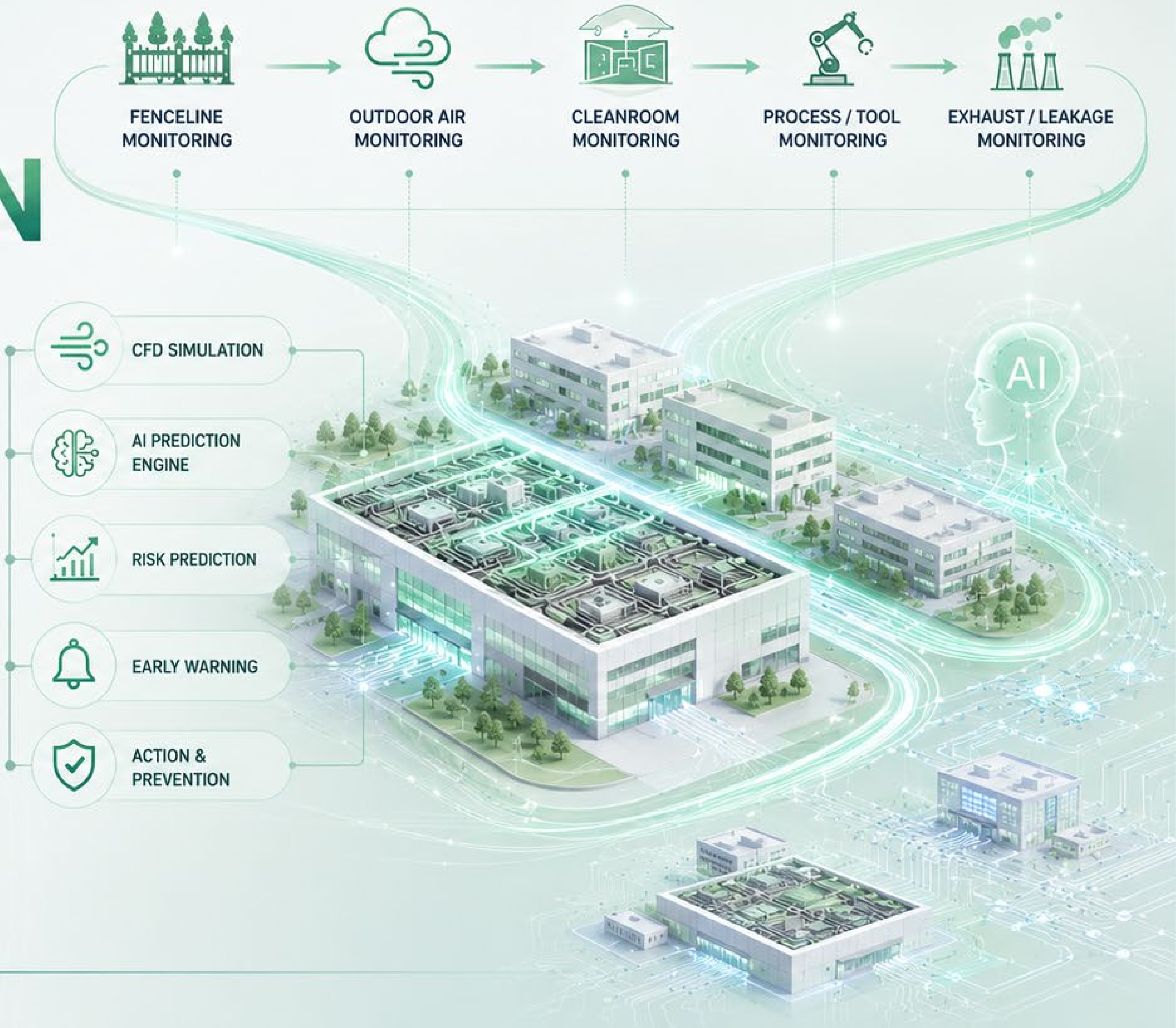
From Passive AMC Monitoring
to Predictive Environmental Intelligence

MONITOR
Detect

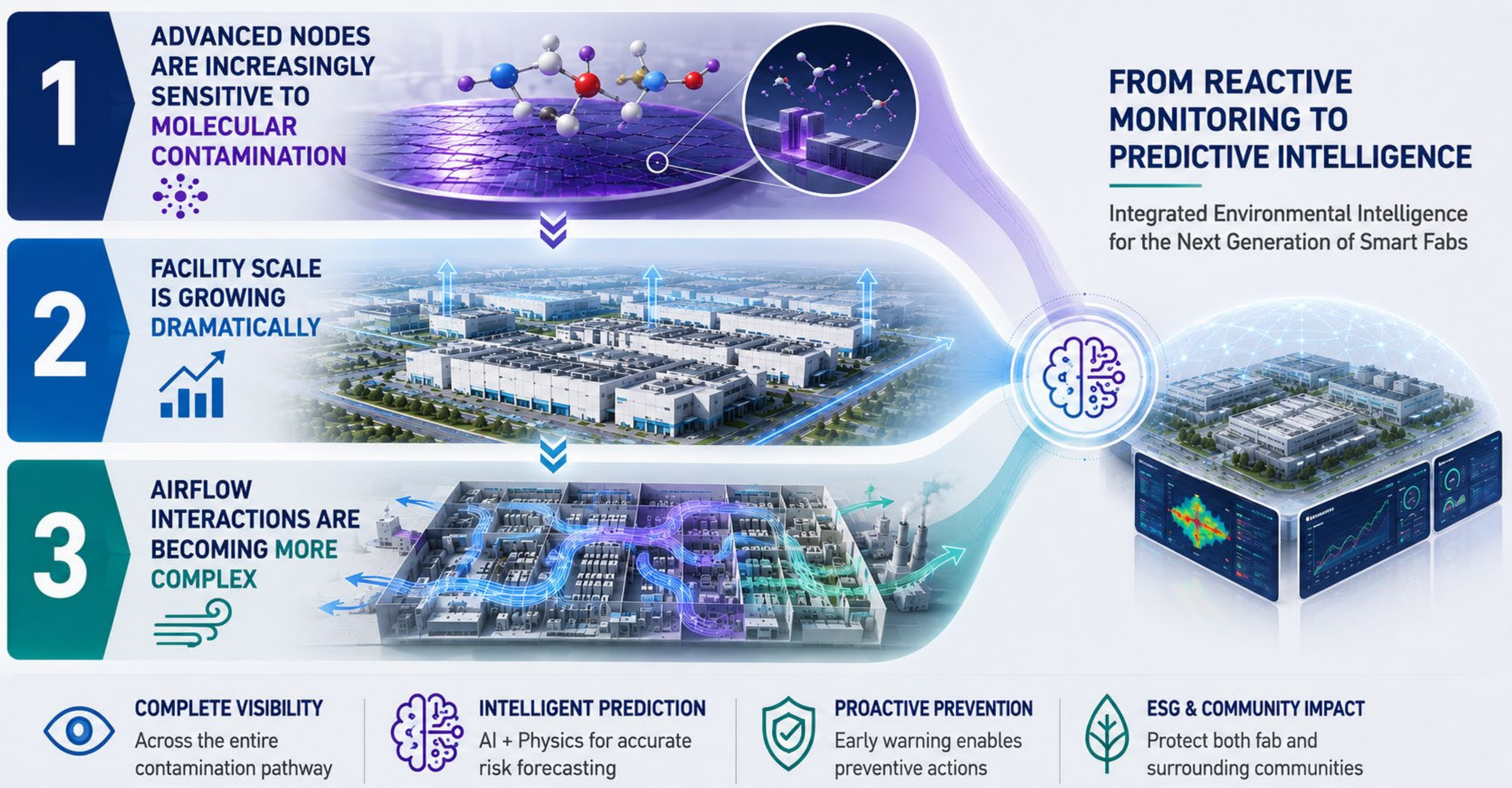
UNDERSTAND
Simulate

PREDICT
Anticipate

PREVENT
Protect



Semi Fab Environmental Risk – AMC Management



AMC Management – Common Practice

Physics-constrained AI for Fab AMC Management

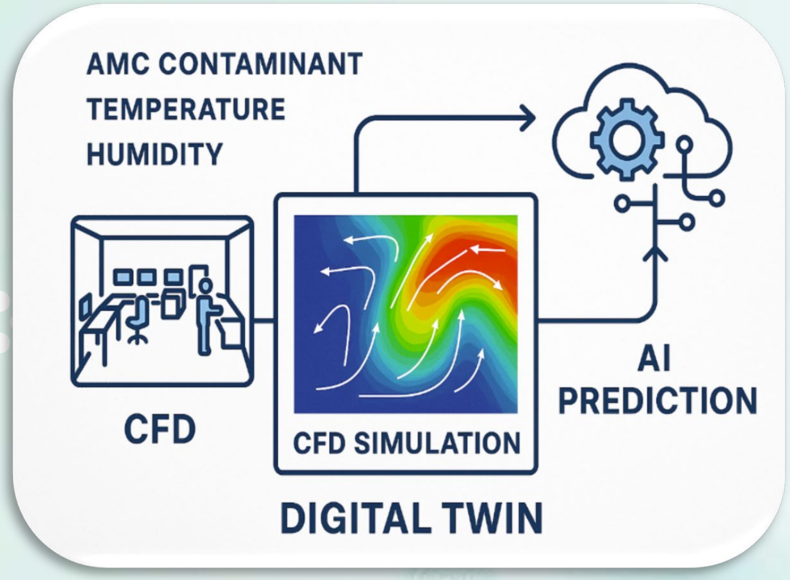
Vision: Environmental Intelligence Platform

From

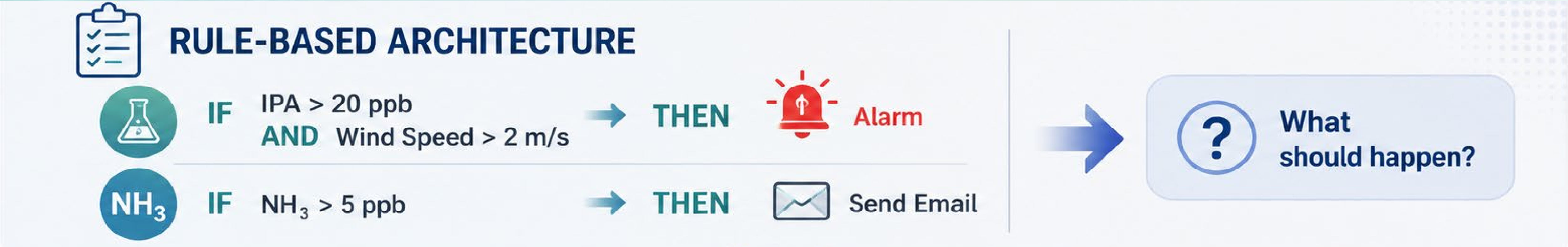
- Sensors/ Analyzers
- 24/7 Monitoring
- Alarm & React

To

- Sensor Network
- Facility Digital Twin
- CFD Airflow Model
- AI Prediction Engine
- Real-time Risk Dashboard
- Preventive Action System



Why Physics-constrained AI?



Step 1- Establish Monitoring Network

Sampling Points

End-to-End Monitoring Coverage

Fenceline

↓↑

MAU / Outdoor Air

↓↑

Cleanroom

↓↑

Process Tool

↓↑

Exhaust / Leakage Source

A unified monitoring strategy across the entire contamination pathway.

Step 2- Data Integration

DATA FLOW

- Facility FMCS
- ↓ Tool FDC
- ↓ Tool Sensor
- ↓ SCADA
- ↓ Weather Station
- ↓ AMC Monitoring
- ↓ Digital Twin

DATA SOURCE EXAMPLES

- Existing AMC monitoring system
- Facility BMS
- HVAC operating conditions
- Weather station
  - Temperature
  - Humidity
  - Wind speed
  - Wind direction
  - Rainfall
- Tool status
  - Operation status
  - Run time
  - Alarm codes
- Alarm history
- Maintenance records

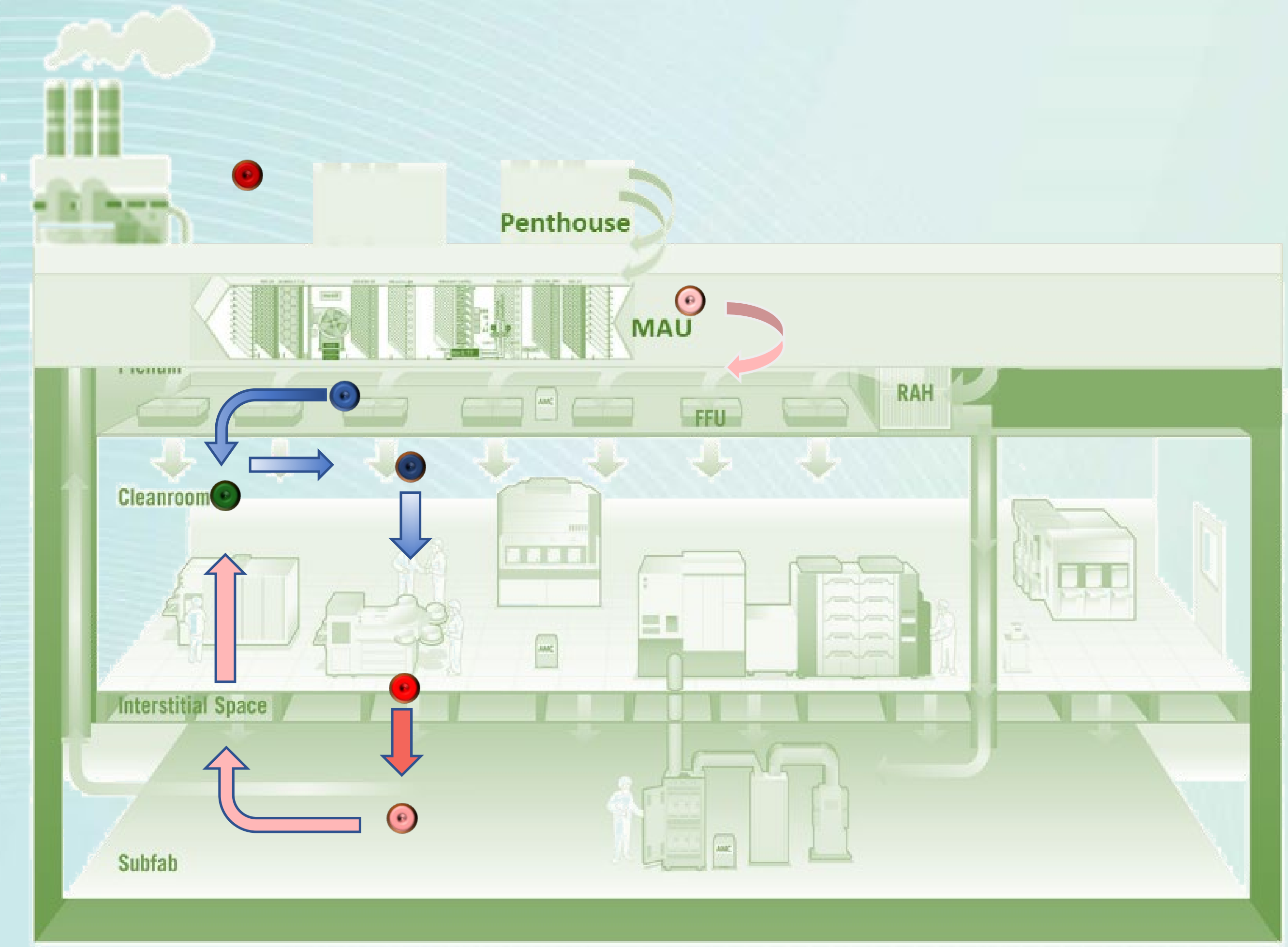
DIGITAL TWIN

AS A LAYER OF DATA INTEGRATION

Integrate	Standardize	Contextualize	Visualize
Multi-source Data	Data Model & Time Sync	Process & Environment	Real-time & Historical View

Unified View • Real-time Insight • Predict & Prevent

Step 3- Digital Twin Establishment – CFD Simulation (1/2)



Sampling Points in Plant

Action:

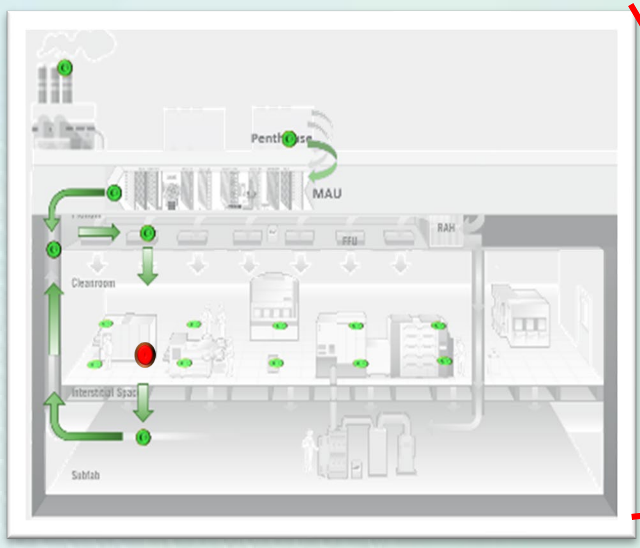
Incorporate Multidimensional Data

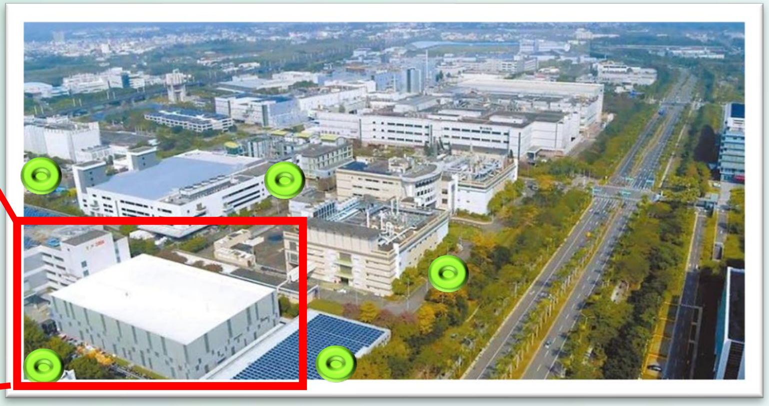
Purpose: Develop a DT for Simulation of Contamination Airflow

Step 3- Digital Twin Establishment – CFD Simulation (2/2)

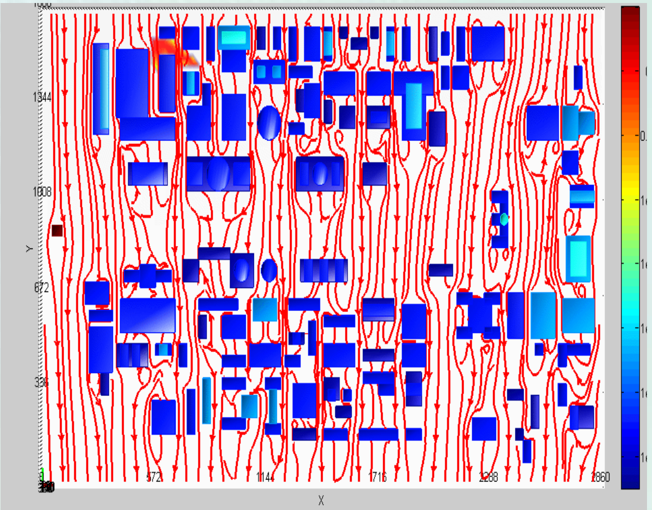
Action:

1. **Identify the Subject (AMC/Particle/Temp./Humidity, etc.)**





2. **Tap in Specific Factors (Diffusion coefficient/Seasonal variables, etc.)**



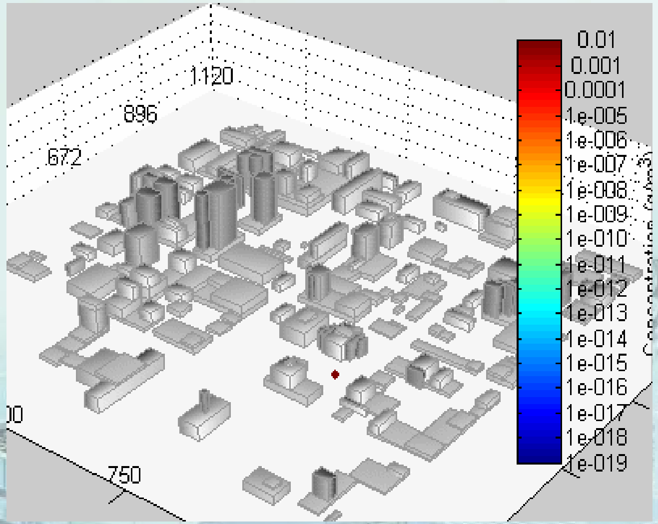
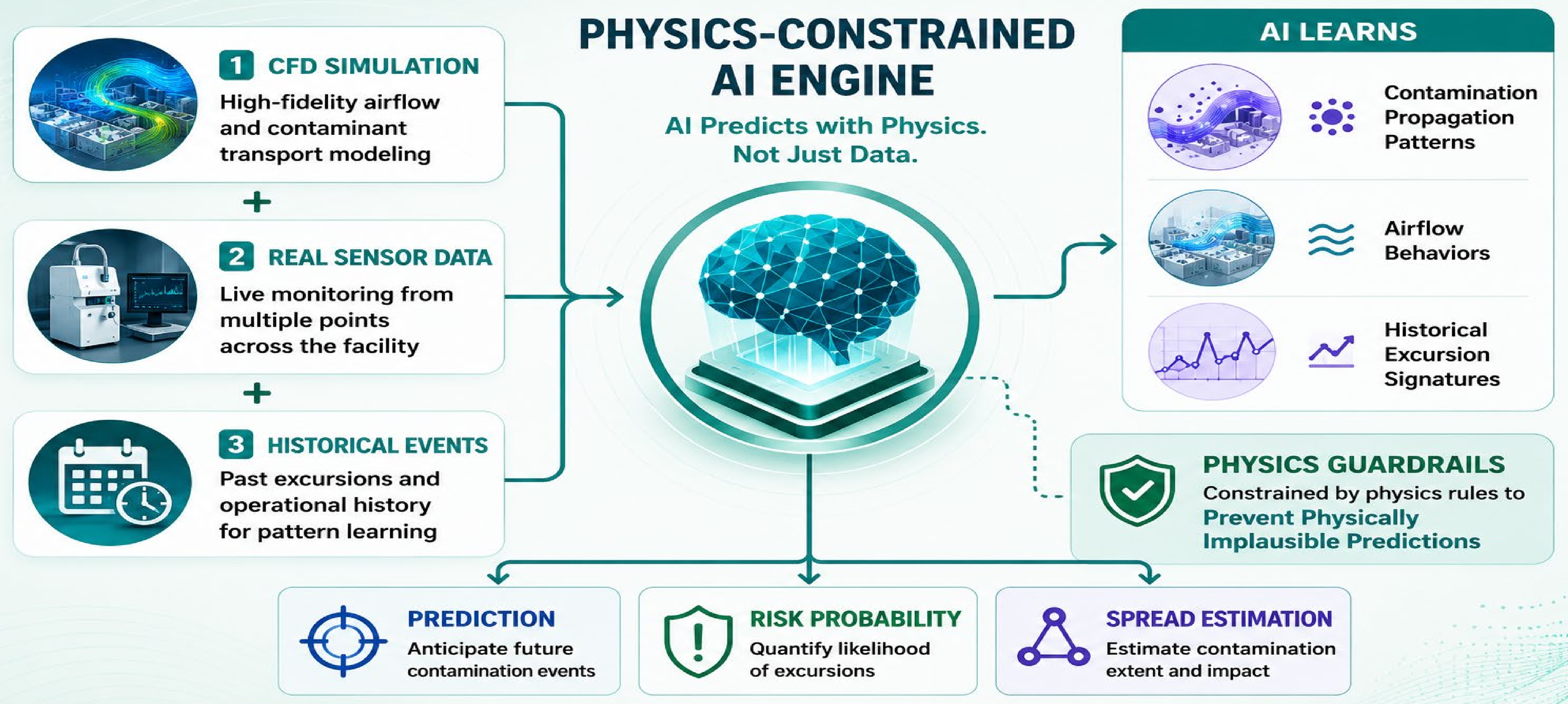


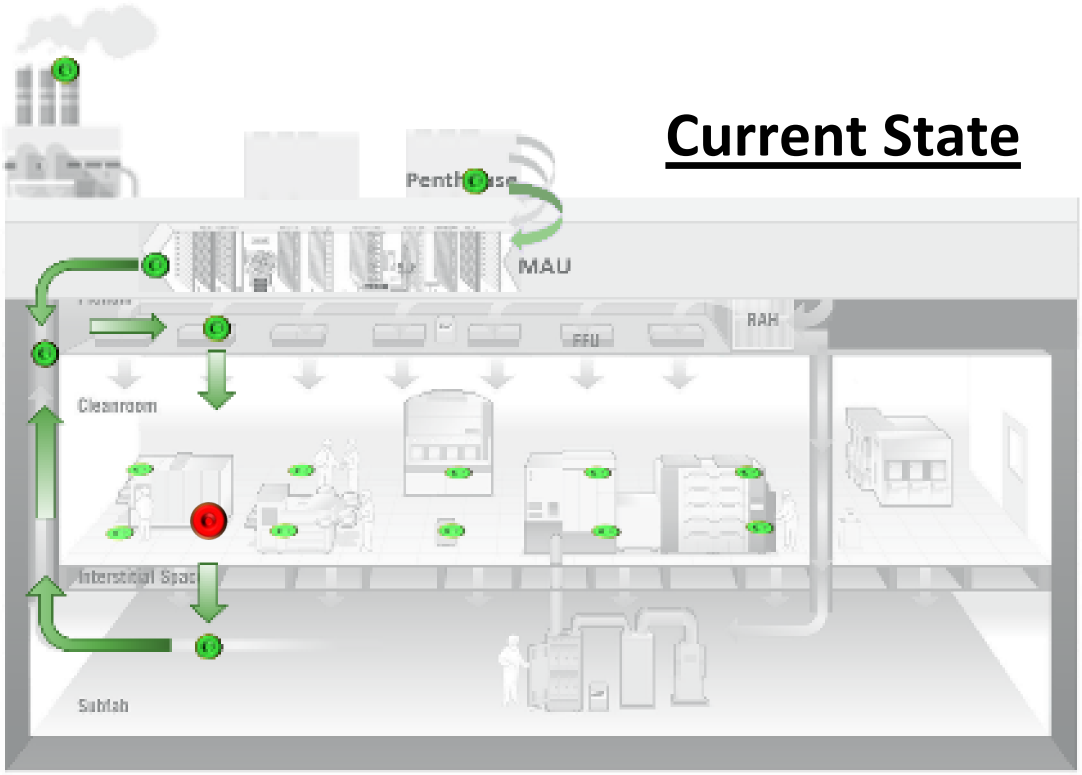
Illustration of Dispersion Modeling

Step 4- Utilize Physics-Constrained AI Solution

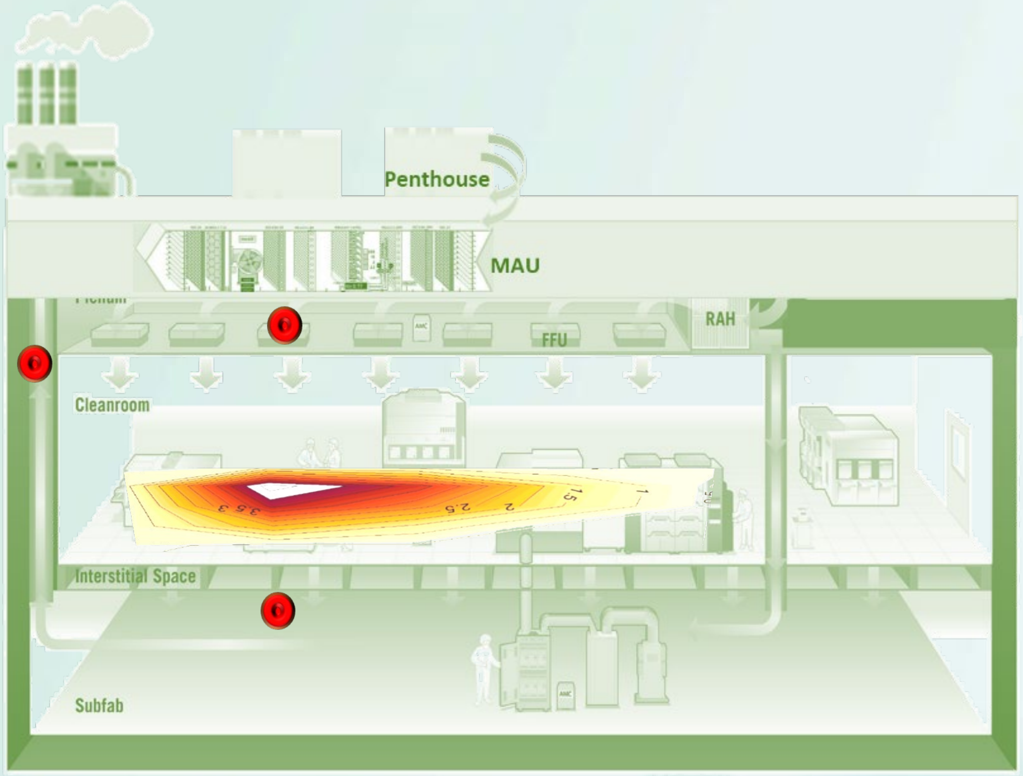


Step 4- Environment Risk Prediction Enabled by AI

Current State



Prediction



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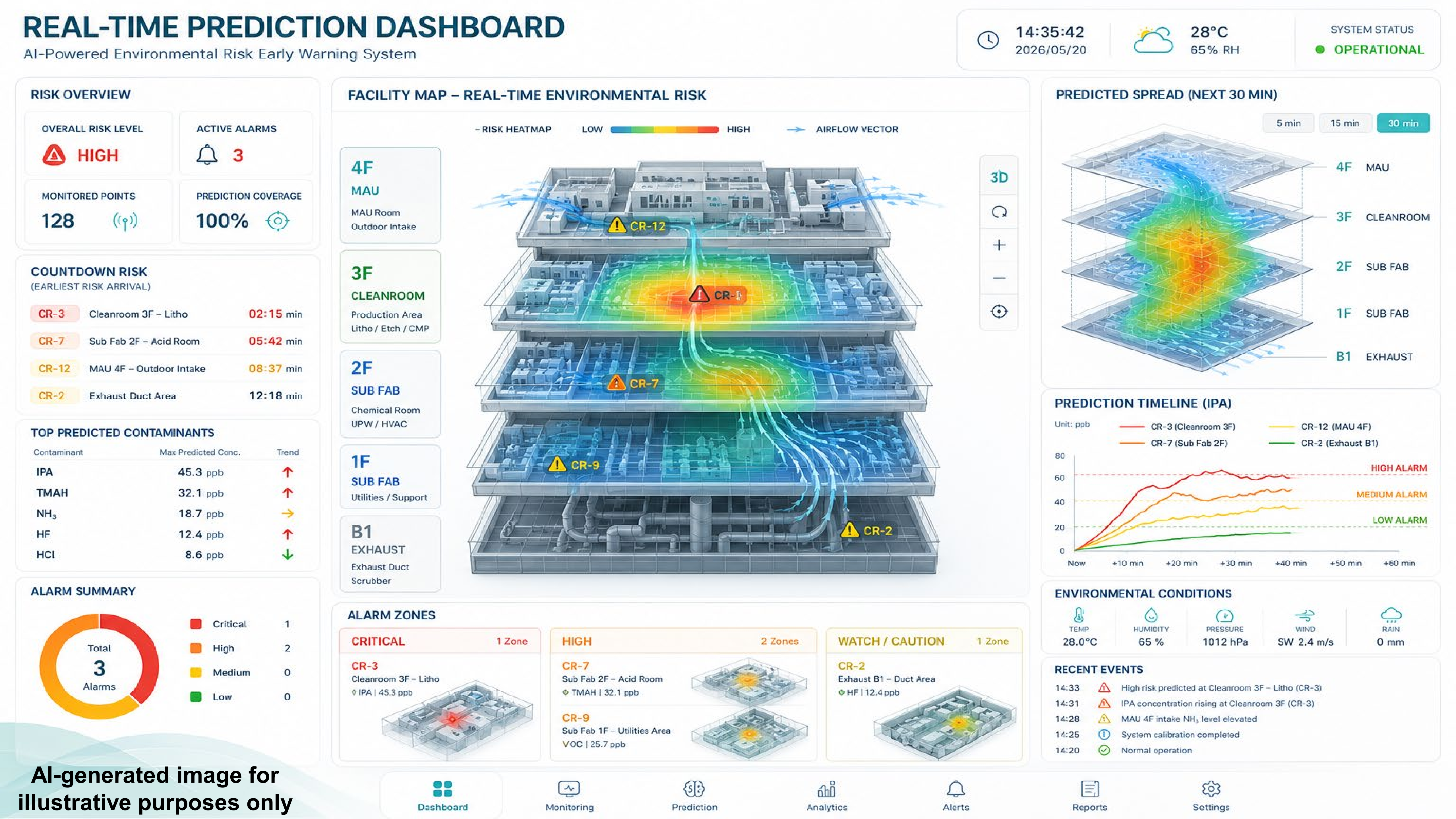

AI-generated image for illustrative purposes only

Real Case Example

Example Scenario: IPA Leakage Scenario

Upon IPA leakage

Conventional Practice/System:

1.alarm after concentration rises

TRICORNTECH's Physics-Constrained AI Digital Twin:

1. Predicts spread direction
2. Estimates affected zones
3. Foresee concentration trend
4. triggers early warning

Scaling – Inside Out → Outside In

- Air travels from inside of fab to outdoor environment and vice versa.
- Adjacent facilities/communities can be the exposed receptor as well as the source of contaminants to the semi. manufacturing.

Value to Smart Facilities

Monitoring

Prediction

Prevention

Key Benefits

- Yield Protection
- Faster Root Cause Analysis
- Reduced Downtime
- Lower Chemical Risk
- Reduced Human Dependency
- Faster Incident Response

ESG-Driven Env. Intelligence for Semiconductor Fabs

ESG-driven

Shift from compliance monitoring to ESG-driven environmental accountability — fabs are now expected to proactively *prevent* AMC pollution (fab + surrounding communities) as part of environmental justice commitments.

AI-Integrated

Integrate AMC sensing with facility-wide airflow simulation and AI forecasting — enabling fabs to correlate process emissions with airflow behavior, worker exposure zones & external impact — unlocking "predict–prevent–prove" strategy.

Sustainability-Embedded

Position AMC control as part of long-term ESG value & carbon / waste reduction — embedding sustainable design and intelligent control into fab operations to reduce environmental impact while maintaining yield competitiveness.

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THANK YOU

BUILDING PREDICTIVE ENVIRONMENTAL INTELLIGENCE FOR A SAFER TOMORROW

MONITOR Detect › **UNDERSTAND** Simulate › **PREDICT** Anticipate › **PREVENT** Protect

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